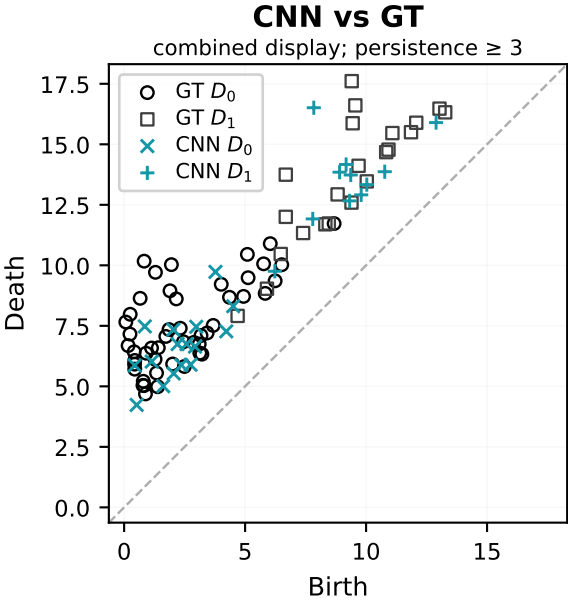
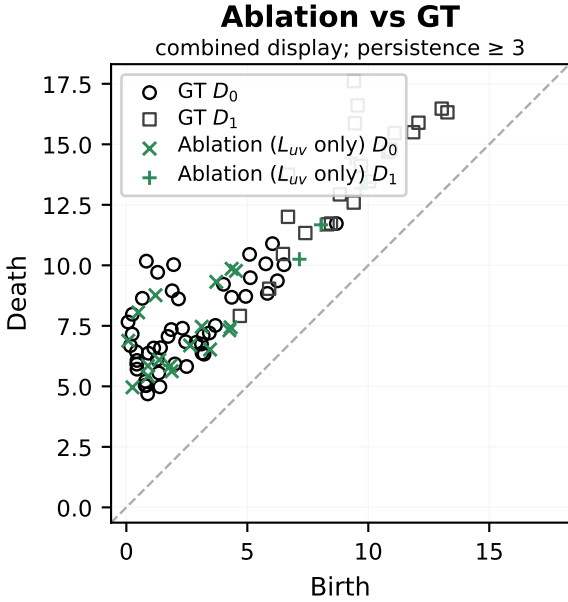
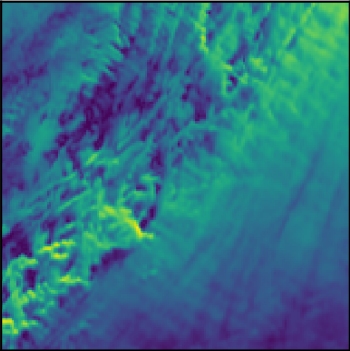


Sample 71: Topology-Aware Gains Beyond Matched Fine-Tuning

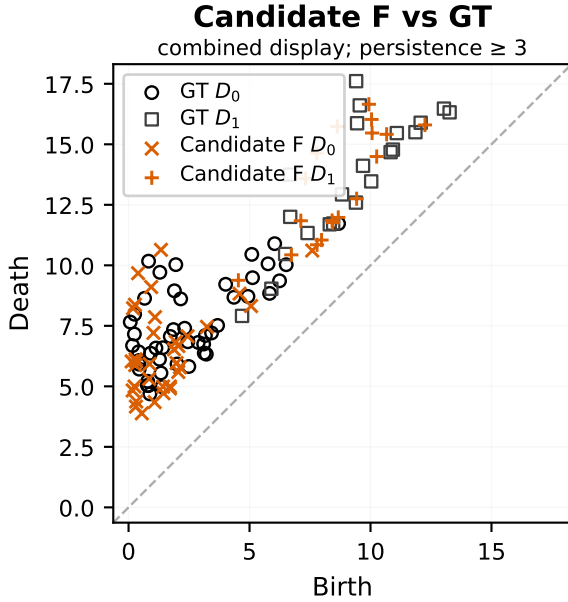
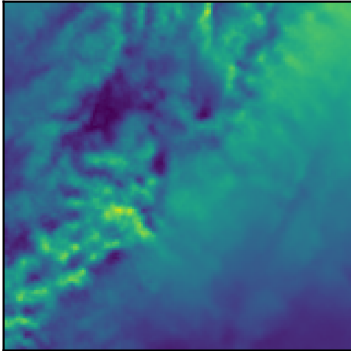
CNN = pretrained baseline; Ablation = matched L_{uv} -only fine-tuning; Candidate F = $L_{uv} + L_{grad} +$ repaired E2 supervision.



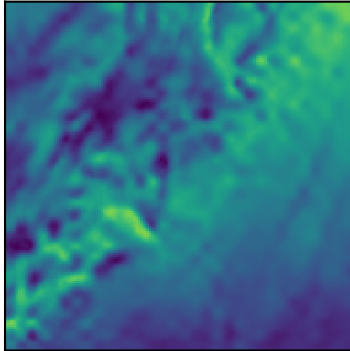
GT



CNN

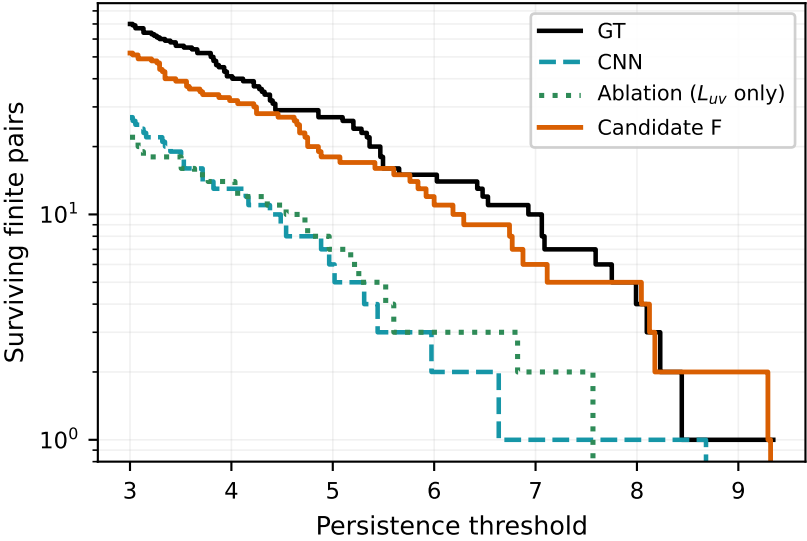


Ablation

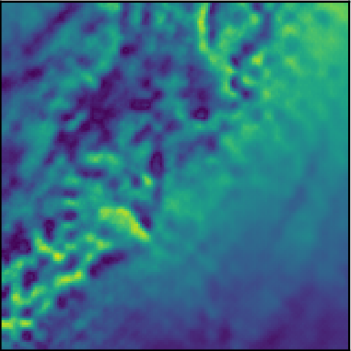


Persistence Survival

combined finite D_0+D_1 ; thresholds ≥ 3

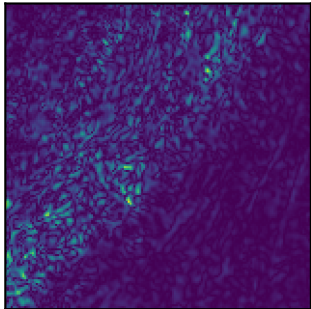


Candidate F

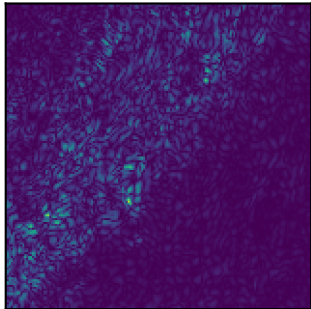


Wind-speed magnitude

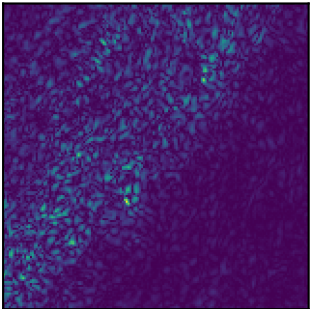
|CNN - GT|



|Ablation - GT|



|Candidate F - GT|



Absolute speed error

Topology chain (lower is better)

$W_{2,\infty}$: CNN 17.140 → Abl. 18.569 → F 11.670

$W_{2,2}$: CNN 21.906 → Abl. 24.000 → F 14.469

Conventional fidelity gaps

F vs CNN (original closeness rank 8/168): $|\Delta\text{PSNR}|=0.606$ dB, MAE=8.6%, RMSE=3.3%

F vs L_{uv} ablation: $|\Delta\text{PSNR}|=1.373$ dB, MAE=18.3%, RMSE=17.4%

Matched-control interpretation

Candidate F is lower than the matched L_{uv} -only fine-tuning control under all three validated PD metrics. This sample therefore supports an auxiliary-loss effect beyond ordinary fine-tuning.